CS 255 Business Requirements Document Template

Complete this template by replacing the bracketed text with the relevant information.

This template lays out all the different sections that you need to complete for Project One. Each section has guiding questions to prompt your thinking. These questions are meant to guide your initial responses to each area. You are encouraged to go beyond these questions using what you have learned in your readings. You will need to continually reference the interview transcript as you work to make sure that you are addressing your client's needs. There is no required length for the final document. Instead, the goal is to complete each section based on your client's needs.

Tip: You should respond in a bulleted list for each section. This will make your thoughts easier to reference when you move into the design phase for Project Two. One starter bullet has been provided for you in each section, but you will need to add more.

System Components and Design

Purpose

What is the purpose of this project? Who is the client and what do they want their system to be able to do?

- The purpose of the project is to create an online platform for new drivers to sign up for scheduled driving lessons as well as access practice materials for DMV driving tests. The client is DriverPass, and on the admin end of the platform DriverPass wants to be able to download user data and reports, reset passwords, block lower admin access, track, view and enter data related to users and trainers, add/remove packages offered by the company, and connect to the DMV website. On the user end of the platform, they want to allow users to sign up for accounts, choose lesson packages, enter user information, schedule driving lessons, access practice materials, and view lesson details and progress.

System Background

What does DriverPass want the system to do? What is the problem they want to fix? What are the different components needed for this system?

- DriverPass wants a system that is an all-in-one platform where new drivers have access to study materials and practice tests as well as scheduled driving lessons. The company is trying to provide new drivers with better driver training. The components of the system are the user (including trainers) accounts, practice materials, driving lesson scheduling, package selection, and database.

Objectives and Goals

What should this system be able to do when it is completed? What measurable tasks need to be included in the system design to achieve this?

- The system should be able to give users access to DMV test practice materials, schedule driving lessons and view lesson details, view training progress, view trainers, and access regulations set by the DMV. The system needs user management, DMV practice test modules, practice session

management, scheduling, progress tracking, trainer management, DMV regulations access, performance tracking, and security.

Requirements

Nonfunctional Requirements

In this section, you will detail the different nonfunctional requirements for the DriverPass system. You will need to think about the different things that the system needs to function properly.

Performance Requirements

What environments (web-based, application, etc.) does this system need to run in? How fast should the system run? How often should the system be updated?

- The system will run in a web-based environment and should load within 2 seconds under ideal conditions. Content (like videos and training modules) should load at the rate that users' internet connection speed allows. The system should be updated monthly or as frequently as needed.

Platform Constraints

What platforms (Windows, Unix, etc.) should the system run on? Does the back end require any tools, such as a database, to support this application?

- The system should run on Windows, Linux, macOS, iOS and Android platforms while also being compatible with the web browsers commonly found on these platforms. The back end will require a database and web server to store account information and host the system, respectively.

Accuracy and Precision

How will you distinguish between different users? Is the input case-sensitive? When should the system inform the admin of a problem?

- The system will differentiate between users based on individualized login credentials and role-based permissions. Login inputs will be case-sensitive to bolster security, and the admin should immediately be informed of problems related to security breaches or user account concerns.

Adaptability

Can you make changes to the user (add/remove/modify) without changing code? How will the system adapt to platform updates? What type of access does the IT admin need?

- Changes to users (such as add/remove/modify) can be made at the user interface level without any changes having to be made to the code. The system will seamlessly adapt to platform updates by continuing to run as long as the platform remains compatible with the current version of the system post update. IT admin require full access to the system including database and backend access to modify/manage stored data as well as adding/removing modules.

Security

What is required for the user to log in? How can you secure the connection or the data exchange between the client and the server? What should happen to the account if there is a "brute force" hacking attempt? What happens if the user forgets their password?

- Usernames and passwords will be required for logging in, and data flowing between the client and server will be encrypted to keep it secure. Brute force hacking attempts made on accounts should notify IT admin and lock the respective account. In the case that a user forgets their password, they can reset it via an admin request or password reset through email.

Functional Requirements

Using the information from the scenario, think about the different functions the system needs to provide. Each of your bullets should start with "The system shall . . ." For example, one functional requirement might be, "The system shall validate user credentials when logging in."

- The system will implement role-based access control.
- The system will create user accounts and store user information.
- The system will always allow access if users are connected to the internet.
- The system will have two factor authentication for logins.
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- The system will track the start and end times for driving lessons.

User Interface

What are the needs of the interface? Who are the different users for this interface? What will each user need to be able to do through the interface? How will the user interact with the interface (mobile, browser, etc.)?

- Users navigating the system on a standard computer browser will interact with the interface through selectable web pages, while mobile users will interact with a touch-based interface.
- The student interface requires a login/account creation section, a main menu that displays online test progress, user information, driver notes, driver photo, user photo, and special needs information. The teacher interface requires the same information as the student interface with the addition of an availability module. And the admin interface requires modules for data tracking, user details, requests, and security alerts.
- The different users are students, teachers, and administrators.

Assumptions

What things were not specifically addressed in your design above? What assumptions are you making in your design about the users or the technology they have?

- Users have access to the internet.
- User devices meet the minimum requirements to run the system on their respective platforms.
- Users possess sufficient technical proficiency to navigate the system successfully.
- Student users will sign up for driving lessons or online training.

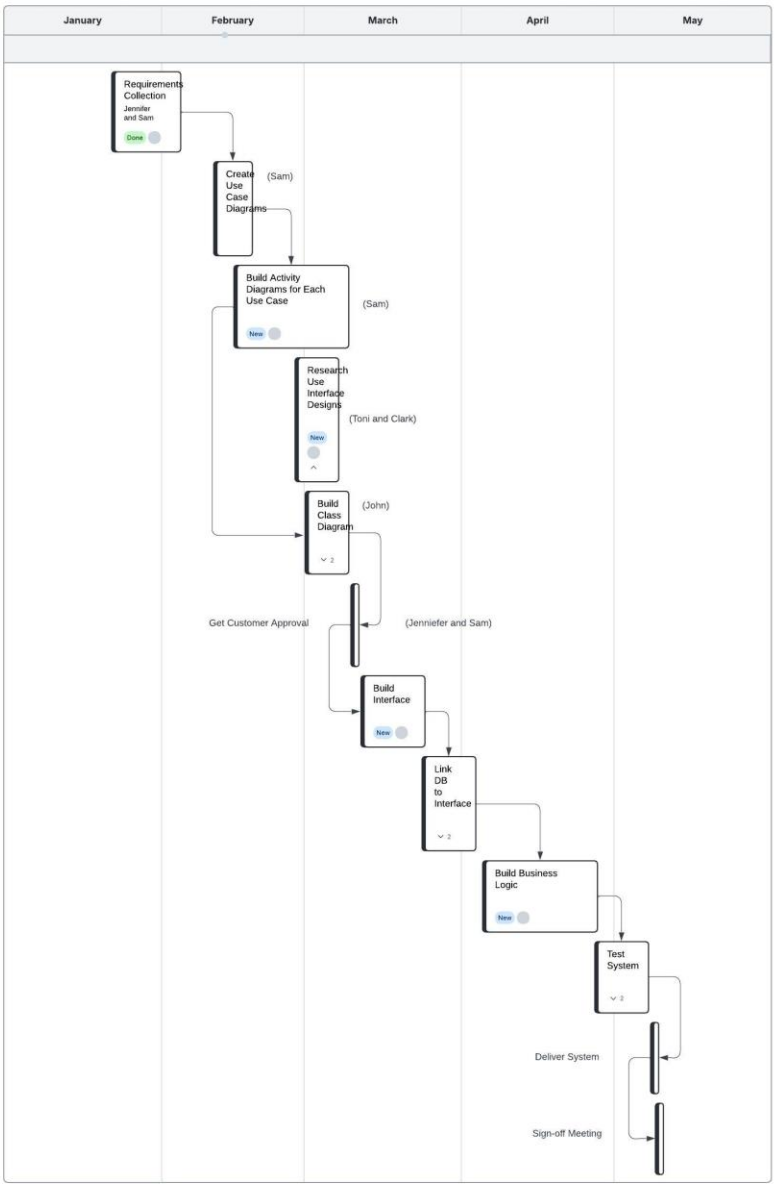
Limitations

Any system you build will naturally have limitations. What limitations do you see in your system design? What limitations do you have as far as resources, time, budget, or technology?

- Resources: Under current circumstances, DriverPass will only be available to users within the United States due to its adherence to DMV regulations.
- Resources: At max capacity there may be up to a 2-hour window between driving lessons if all 10 cars are in use.
- Technology: The system needs to perform successfully across various platforms while offering users standardized experience (quick loading, minimal errors).
- Technology: Server outages/system downtime may hinder driving lesson scheduling.
- Time: The system must be completed and delivered by the proposed project end date.

Gantt Chart

Please include a screenshot of the GANTT chart that you created with Lucidchart. Be sure to check that it meets the plan described by the characters in the interview.


January
February
March
April
May
Requirements Collection
Jennifer and Sam
Done
Create Use Case Diagrams
(Sam)
Build Activity Diagrams for Each Use Case
(Sam)
New
Research Use Interface Designs
(Toni and Clark)
New
Build Class Diagram
(John)
Get Customer Approval
(Jenniefer and Sam)
Build Interface
New
Link DB to Interface
Build Business Logic
New
Test System
Deliver System
Sign-off Meeting

Functional requirements: DriverPass will need to connect to the DMV's the website and display training progress on an individual basis. Connecting to the DMV website will allow students to stay up-to-date with the latest rules and regulations provided by the DMV. Additionally, displaying training progress gives students the ability to track their advancement through the training material, pinpoint which subjects require more study, and see how well they understand the information.
Nonfunctional requirements: While not as vital to DriverPass' success as the requirements listed above, it's important that the system runs on various operating systems (Windows, Linux, macOS, iOS, and Android) and runs in a web-based environment. Being compatible with multiple operating systems gives DriverPass access to a wider market of users and ensures that those users can get on the app with no issues. Running DriverPass in a web-based environment provides users with the application without having to download anything.

> The use case diagram essentially represents the DriverPass app and the different kinds of users that will interact with it. The stick figures outside of the box are students, teachers, and administrative users. The circles broadly represent things that users can do in the app and the lines that extend from the users to the circles show which users are related to certain activities. Students can do things like manage their accounts and make appointments, teachers can manage their schedules, and administrators can access all of these activities in the background to fix issues or alter information.

This use case breaks down the process of a student navigating through the "manage appointments" section of the app. Initially, the system verifies that the student has an appointment and allows them to view the details, cancel the appointment, or change the details if an appointment exists. If no appointment exists, the student can make one. This functionality is crucial because it's where students will schedule their driving lessons.

Prioritization of security standards is a critical step in designing a safe and dependable application. DriverPass will implement multifactor authentication to provide an extra layer of security by verifying account access through a phone number, email, or 3rd party authenticator. The data that's shared between the client and server will be encrypted to prevent sensitive information from falling into the wrong hands. Different kinds of users will have different sorts of privileges with respect to the application's full set of controls/capabilities. For example, students won't be able create availability schedules or make notes on driving lessons. Finally, accounts will lock users out after multiple failed login attempts, and administrators will have activity logs and alerts to oversee account concerns and potential security breaches.

There are certain limitations that DriverPass may encounter due to uncontrollable factors that outside the scope of the application. DriverPass may experience diminished performance during high-traffic periods if the server capacity is exceed or scaling is limited. Older devices or outdated browsers may affect user experience due to the minimum requirements for hardware and software needed for optimal system performance. Furthermore, integration with external

services (DMV data or email systems) may be affected by 3rd party outages or other adverse situations that are beyond the system's control.